UNIFIED SOIL CLASSIFICATION
INCLUDING IDENTIFICATION AND DESCRIPTION

FIELD IDENTIFICATION PROCEDURES (Excluding particles larger than 3 inches and basing fractions on estimated weights)				GROUP SYMBOLS	TYPICAL NAMES	INFORMATION REQUIRED FOR DESCRIBING SOILS
COARSE GRAINED SOILS — More than half of material is larger than No. 200 sieve size 12 — (The No. 200 sieve size is about the smallest particle visible to the naked eye)	GRAVELS — More than half of coarse fraction is larger than No. 4 sieve size — (For visual classifications, the ¼" size may be used as equivalent to the No. 4 sieve size)	CLEAN GRAVELS (Little or no fines)	Wide range in grain size and substantial amounts of all intermediate particle sizes.	GW	Well graded gravel's, gravel–sand mixtures; little or no fines	Give typical name; indicate approximate percentages of sand and gravel; max. size; angularity, surface condition, and hardness of the course grains; local or geologic name and other pertinent descriptive information, and symbol in parentheses For undisturbed soils add information on stratification, degree of compactness, cementation, moisture conditions and drainage characteristics. EXAMPLE:– Silty Sand, gravely; about 20% hard, angular gravel particles ½ – inch maximum size; rounded and subangular sand grains coarse to fine; about 15% non–plastic fines with low dry strength; well compacted and moist in place; alluvial sand; (SM)
			Predominantly one size or a range of sizes with some intermediate sizes missing.	GP	Poorly graded gravel's, gravel–sand mixtures; little or no fines	
		GRAVELS WITH FINES (Appreciable amount of fines)	Non–plastic fines (for identification procedures see ML below).	GM	Silty gravels, poorly graded gravel–sand–silt mixtures	
			Plastic fines (for identification procedures see CL below).	GC	Clayey gravels, poorly graded gravel–sand–clay mixtures	
	SANDS — More than half of coarse fraction is smaller than No. 4 sieve size	CLEAN SANDS (Little or no fines)	Wide range in grain sizes and substantial amounts of all intermediate particle sizes.	SW	Well graded sands, gravely sands; little or no fines	
			Predominantly one size or a range of sizes with some intermediate sizes missing.	SP	Poorly graded sands, gravely sands; little or no fines	
		SANDS WITH FINES (Appreciable amount of fines)	Non–plastic fines (for indentification procedures see ML below).	SM	Silty sands, poorly graded sand–silt mixtures	
			Plastic fines (for identification procedures see CL below).	SC	Clayey sands, poorly graded sand–clay mixtures	

IDENTIFICATION PROCEDURES ON FRACTION SMALLER THAN No 40 SIEVE SIZE						
		DRY STRENGTH (Crushing Characteristics)	DILATANCY (Reaction to skaking)	TOUGHNESS (Consistency near Plastic Limit)		
FINE GRAINED SOILS — More than half of material is smaller than No. 200 sieve size — (The No. 200 sieve size is about the smallest particle visible to the naked eye)	SILTS AND CLAYS — Liquid limit less than 50	None to slight	Quick to slow	None	ML	Inorganic silty and very fine sands, rock flour, silty or clayey fine sands with slight plasticity
		Medium to high	None to very slow	Medium	CL	Inorganic clays to low to medium plasticity, gravely clays, sandy clays, silty clays, lean clays
		Slight to medium	Slow	Slight	OL	Organic silts and organic silt–clays of low plasticity
	SILTS AND CLAYS — Liquid limit greater than 50	Slight to medium	Slow to none	Slight to medium	MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts
		High to very high	None	High	CH	Inorganic clays of high plasticity, fat clays
		Medium to high	None to very slow	Slight to medium	OH	Inorganic clays of medium to high plasticity
HIGHLY ORGANIC SOILS		Readily identified by color, odor, spongy feel and frequently by fibrous texture.			PT	Peat and other highly organic soils

Give typical name; degree and character of plasticity, amount and maximum size of coarse grains; color in wet condition, odor in any, local or geologic name, and other pertinent descriptive information; and symbol in parentheses.

For undisturbed soils add information on structure, stratification, consistency in undisturbed and remolded states. Moisture and drainage conditions

EXAMPLE:–
Clayey Silt, brown; slightly plastic. small percentage of fine sand. numerous vertical root holes, firm and dry in place, loess, (ML)

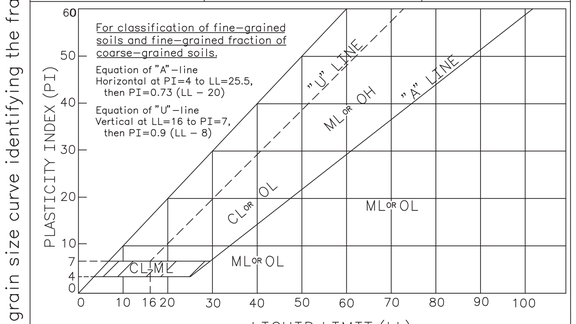
LABORATORY CLASSIFICATION CRITERIA

Use grain size curve identifying the fractions as given under field identification

Determine percentages of gravel and sand from grain size curve. Depending on percentage of fines (fraction smaller than No. 200 sieve size) coarse grained soils are classified as follows:

Less than 5%	GW, GP, SW, SP
More than 12%	GM, GC, SM, SC
5% to 12%	Bordering cases requiring use of dual symbols.

$C_V = \frac{D_{60}}{D_{10}}$ Greater than 4 $C_C = \frac{(D_{30})2}{D_{10}\ Dep}$ Between one and 3	
Not meeting all gradation requirements for GW	
Atterberg limits below "A" line or PI less than 4	Above "A" line with PI between 4 and 7 are borderline cases requiring use of dual symbols.
Atterberg limits above "A" line with PI greater than 7	
$Cv = \frac{D\ 60}{D\ 10}$ Greater than 6 $Cc = \frac{(D\ 30)2}{D\ 10\ Dep}$ Between one and 3	
Not meeting all gradation requirements for SW	
Afterberg limits below "A" line or PI less than 4	Above "A" line with PI between 4 and 7 are borderline cases requiring use of dual symbols.
Afterberg limits above "A" line with PI greater than 7	



FOR LABORATORY CLASSIFICATION OF FINE GRAINED SOILS 110

* Boundary classifications: – Soils processing characteristics of two groups are designated by combinations of group symbols for Example GW–GC, well graded gravel mixture with clay binder
* All sieve sizes on this chart are US standard.

FIELD IDENTIFICATION PROCEDURES FOR FINE GRAINED SOILS OR FRACTIONS

These procedures are to be performed on the minus No. 40 sieve size particles, in for field classification purposes, screening is not intended, simply remove by hand the coarse particles that interfere with the tests

DILATANCY (Reaction to shaking)

After removing particles larger than No 40 sieve sieve, prepare a pat striking vigorously against the other of moist soil with a volume of about one–half cubic inch. Add enough water if necessary to make the soil soft but not sticky.
Place the pot in the open palm of one hand and shake horizontally, hand several times. A positive reaction consists of the appearance of water on the surface of the pot which changes to a livery consistency and becomes glossy. When the sample is squeezed between the fingers, the water and gloss disappear from the surface, the pot stiffens, and finally it cracks or crumbles. The rapidity of appearance of water during shaking and of its disappearance during squeezing assist in identifying the character of the times in a soil.
Very fine clean sands give the quickest and most distinct reaction whereas a plastic clay has no reaction inorganic silts, such as a typical rock flour, show a moderately quick reaction.

DRY STRENGTH (Crushing Characteristics)

After removing particles larger than No. 40 sieve size, mold a pot of soil to the consistency of putty, adding water if necessary. Allow the pot to dry completely by oven, sun, or air drying and then test its strength by breaking and crumbling between the fingers. This strength is a measure of the character and quantity of the colloidal fraction contained in the soil. The dry strength increases with increasing plasticity.
High dry strength is characteristic for clays of the CH group A typical inorganic silt possesses only very slight dry strength Silty fine sands and silts have about the same slight dry strength, but can be distinguished by the feel when powdering the dried specimen. Fine sand feels gritty whereas a typical silt has the smooth feel of flour
Letter symbols in the logs, are group symbols of the Unified soil classification system based on field indent. Copies of Drawing No. 103–D–347, Unified Soil Classification, may be obtained on request office of Director of Design and Construction. Bureau of Reclamation, Denver, Colorado R0225

TOUGHNESS (Consistency near plastic limit)

After removing particles larger than No. 40 sieve size, a specimen of soil about one–half inch cube in size is molded to the consistency of putty. If too dry, water must be added and if sticky, the specimen should be spread out in a thin layer and allowed to lose some moisture by evaporation. Then the specimen is rolled out by hand on a smooth surface or between the palms into a thread about one–eighth inch in diameter. The thread is then folded and rerolled repeatedly. During this manipulation, the moisture content is gradually reduced and the specimen stiffens, finally loses its plasticity, and crumbles when the plastic limit is reached.
After the thread crumbles, the pieces should be lumped together and a slight kneading action continued until the lump crumbles.
The tougher the thread near the plastic limit and the stiffer the lump when it finally crumbles, the more potent is the colloidal clay fraction in the soil weakness of the thread at the plastic limit and quick loss of coherence of the lump below the plastic limit indicate either inorganic clay of low plasticity, or materials such as keolin – type clays and organic clays which occur below the A–line.
Highly organic clays have a very weak and spongy feel at the plastic limit

ADOPTED BY – CORPS OF ENGINEERS AND BUREAU OF RECLAMATION 1952

AS BUILT 11–7–84 116–PCW

DATE AND TIME PLOTTED
Not Plotted
PLOTTED BY
Not Plotted

G1
103–D–347

WEATHERING

FRESH (W1): Body of rock is not oxidized or discolored; fracture surfaces are not oxidized or discolored*; no separation of grain boundaries; no change of texture and no solutioning. Hammer rings when crystalline rocks are struck.

SLIGHTLY WEATHERED TO FRESH (W2):**

SLIGHTLY WEATHERED (W3): Discoloration or oxidation is limited to surface of, or short distance from fractures; some feldspar crystals are dull; fracture surfaces have minor to complete discoloration or oxidation; no visible separation of grain boundaries; texture preserved and minor leaching of soluble minerals may be present. Hammer rings when crystalline rocks are struck, body of rock is not weakened by weathering.

MODERATELY TO SLIGHTLY WEATHERED (W4):**

MODERATELY WEATHERED (W5): Discoloration or oxidation extends from fractures, usually throughout body of rock; ferromagnesian minerals are "rusty", feldspar crystals are "cloudy"; all fracture surfaces are discolored or oxidized; partial opening of grain boundaries visible; texture generally preserved, but soluble minerals may be mostly leached. Hammer does not ring when rock is struck, body of rock is slightly weakened.

INTENSELY TO MODERATELY WEATHERED (W6):**

INTENSELY WEATHERED (W7): Body of rock is discolored or oxidized throughout; all feldspars and ferromagnesian minerals are altered to clay to some extent. All fracture surfaces are discolored or oxidized, and friable; partial separation of grain boundaries, rock is friable; in situ disaggregation of granitics common in semi–arid regions; texture altered and leaching of soluble minerals may be complete. Rock has dull sound when struck with hammer; rock is weakened, usually can be broken with moderate to heavy manual pressure or by light hammer blow without reference to planes of weakness.

VERY INTENSELY WEATHERED (W8):**

DECOMPOSED (W9): Body of rock is discolored or oxidized throughout, but resistant minerals such as quartz may be unaltered; all feldspars and ferro–magnesian minerals are completely altered to clay; complete separation of grain boundaries (disaggregated), partial or complete remnant rock structure may be preserved, but resembles a soil.

NOTE: Weathering categories are established primarily for crystalline rocks and those with ferromagnesian minerals, weathering in various sedimentary rocks will not always fit the categories established – weathering categories may be modified for particular site conditions or alteration such as hydrothermal alteration. Where modified criteria are established, they should be identified and described.

* Characteristics of fracture surfaces do not include directional weathering along shears or faults and their associated fracture zones; for example a shear that carries weathering to great depths in a fresh rock mass would not require the whole rock mass to be classified as weathered.

** Combination descriptors are used when equal distribution of both weathering characteristics are present over significant intervals or where characteristics noted are "in between" the diagnostic characteristics.

DURABILITY INDEX

DURABILITY DESCRIPTOR	DESCRIPTIVE CRITERIA
DIØ	Rock specimen or exposure remains intact with no deleterious cracking after exposure longer than 1 year.
DI1	Rock specimen or exposure develops hairline cracking on surfaces within 1 month, but no disaggregation within 1 year of exposure.
DI2	Rock specimen or exposure develops hairline cracking on surfaces within 1 week, and/or disaggregation within 1 month of exposure.
DI3	Specimen or exposure may develop hairline cracks in 1 day and displays pronounced separation of bedding and/or disaggregation within 1 week of exposure.
DI4	Specimen or exposure displays pronounced cracking and disaggregation within 1 day (24 hours) of exposure. Generally ravels and degrades to small fragments.

COLOR

The Munsell color system (Geologic Society of America Rock Color Chart) should be used. This system defines wet color by its hue, value, and chroma. Color symbols used (i.e., 5 YR 5/6 may be included).

SEDIMENTARY AND PYROCLASTIC ROCK PARTICLE SIZES

Size in mm	Sedimentary Rounded, subrounded, subangular		Pyroclastic	
	Particle or fragment	Lithified product	Fragment	Lithified product
256	Boulder	Boulder conglomerate	Block[a] or Bomb[b]	Volcanic[a] breccia or Volcanic[b] agglomerate
64	Cobble	Cobble conglomerate		
4	Pebble	Pebble conglomerate	Lapilli	Lapillistone and Lapilli tuff
2	Granule	Granule conglomerate		
1	Very coarse sand	Sandstone (Very coarse, coarse, medium, fine, or very fine)	Coarse ash	Coarse tuff
0.5	Coarse sand			
0.25	Medium sand			
0.125	Fine sand			
0.0625	Very fine sand			
0.00391	Silt	Siltstone/ Shale	Fine ash	Fine tuff
	Clay	Claystone Shale		

[a] Broken from previous igneous rock, block shaped (angular to subangular).
[b] Solidified from plastic material while in flight, rounded clasts.

IGNEOUS AND METAMORPHIC ROCK TEXTURE

TEXTURE DESCRIPTOR	AVERAGE GRAIN DIAMETER
VERY COARSE GRAINED OR PEGMATITIC	>10 mm [>3/8 in]
COARSE GRAINED	5–10 mm [3/16 –3/8 in]
MEDIUM GRAINED	1–5 mm [1/32 –3/16 in]
FINE GRAINED	0.1–1 mm [0.004 – 1/32 in]
APHANITIC (Cannot be seen with the unaided eye)	<0.1 mm [<0.004 in]

ADDITIONAL TEXTURAL ADJECTIVES

PIT (pitted) – pinhole to 0.03 ft [3/8 in] (<1 to 10 mm) openings.

VUG (vuggy) – Small openings (usually lined with crystals) ranging in diameter from 0.03 ft [3/8 in] to 0.33 ft [4 in] (10 to 100 mm).

CAVITY – An opening larger than 0.33 ft [4 in] (100 mm), size descriptions are required, and adjectives such as small, large, etc., may be used.

HONEYCOMBED – If numerous enough that only thin walls separate individual pits or vugs, this term further describes the preceding nomenclature to indicate cell–like form.

VESICLE (vesicular) – Small openings in volcanic rocks of variable shape and size formed by entrapped gas bubbles during solidification.

BEDDING FOLIATION OR FLOW TEXTURE

DESCRIPTORS	THICKNESS/SPACING
MASSIVE	Greater than 10 ft (>3 m)
VERY THICKLY (bedded, foliated or banded)	3 to 10 ft (1 to 3 m)
THICKLY	1 to 3 ft (300 mm to 1 m)
MODERATELY	0.3 to 1 ft (100 to 300 mm)
THINLY	0.1 to 0.3 ft (30 to 100 mm)
VERY THINLY	0.03 [3/8 in] to 0.1 ft (10 to 30 mm)
LAMINATED (Intensely foliated or banded)	Less than 0.03 ft [3/8 in] (<10 mm)

BEDROCK HARDNESS/STRENGTH

EXTREMELY HARD (H1): Core, fragment or exposure cannot be scratched with knife or sharp pick; can only be chipped with repeated heavy hammer blows.

VERY HARD (H2): Cannot be scratched with knife or sharp pick. Core or fragment breaks with repeated heavy hammer blows.

HARD (H3): Can be scratched with knife or sharp pick with difficulty (heavy pressure). Heavy hammer blow required to break specimen.

MODERATELY HARD (H4): Can be scratched with knife or sharp pick with light or moderate pressure. Core or fragment breaks with moderate hammer blow.

MODERATELY SOFT (H5): Can be grooved 1/16 inch (2 mm) deep by knife or sharp pick with moderate or heavy pressure. Core or fragment breaks with light hammer blow or heavy manual pressure.

SOFT (H6): Can be grooved or gouged easily by knife or sharp pick with light pressure, can be scratched with fingernail. Breaks with light to moderate manual pressure.

VERY SOFT (H7): Can be readily indented, grooved or gouged with fingernail, or carved with a knife. Breaks with light manual pressure.

Any bedrock unit softer than H7, Very Soft, is to be described using USBR 5005–86 (visual classification of soils) consistency characteristics.

REV NO		
1	3–8–00 D – P. M. R.	CONVERTED ORIGINAL DRAWING 40–D–6493 TO ACAD, CHANGED DWG. NO., MINOR REVISIONS.

ALWAYS THINK SAFETY

UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

GEOLOGY FOR DESIGN & SPECIFICATIONS

STANDARD DESCRIPTORS AND DESCRIPTIVE CRITERIA FOR ROCK

GEOLOGY NOMENCLATURE COMMITTEE CHECKED CHUCK SULLIVAN
DRAWN MARSHALL MONSON TECH. APPROVAL PETER M. ROHRER
APPROVED MARK McKEOWN (PEER REVIEWER)

CADD SYSTEM	CADD FILENAME	DATE AND TIME PLOTTED
AutoCAD Rel. 15.0 DENVER, COLORADO	40–D–7022.DWG MARCH 8, 2000	JUNE 26, 2002 11:28

40–D–7022

DISCONTINUITY TERMINOLOGY

DISCONTINUITY – A collective term used for all structural breaks in geologic materials which usually are unhealed and have zero or low tensile strength. Discontinuities also may be healed and exhibit high tensile strength. Discontinuities comprise fractures (including joints), planes of weakness, shears/faults, and shear/fault zones. Contacts between various units also may be considered discontinuities.

FRACTURE – A term used to describe any natural break in geologic material excluding shears and shear zones. Additional fracture terminology is provided below.

SHEAR – A structural break where differential movement has taken place along a surface or zone of failure by shear; characterized by striations, slickensides, gouge, breccia, mylonite, or any combination of these. Often direction, amount of displacement, and continuity may not be known because of limited exposures or observations.

FAULT – A shear with significant continuity which can be correlated between observations; occurs over a significant portion of a given site, foundation area, or region; or is a segment of a fault or fault zone defined in the literature. The designation of a shear as a fault or fault zone is a site-specific determination.

SHEAR/FAULT ZONE – A shear that is expressed in relative terms of width. The zone may consist of gouge, breccia, or many related faults or shears together with fractured and crushed rock between the shears and faults, or any combination of these. In the literature many fault zones simply are referred to as faults.

SHEAR-/FAULT-DISTURBED ZONE – An associated zone of fractures and/or folds adjacent to a shear or shear zone where the country rock has been subjected to only minor cataclastic action and may be mineralized. If adjacent to a fault or fault zone, the term is fault-disturbed zone. Occurrence, orientation, and areal extent of these phenomena depend upon depth of burial (pressure and temperature) during shearing, brittleness of materials, and the stress envelope.

FRACTURE TERMINOLOGY

EXAMPLES SHOWN FOR CORE, BUT APPLICABLE TO ANY OBSERVATION

JOINT (JT) – A relatively planar fracture along which there has been little or no shearing displacement.



FOLIATION JOINT (FJ) OR BEDDING JOINT (BJ) – a relatively planar fracture which is parallel to foliation or bedding along which there has been little or no shearing displacement.



BEDDING PLANE SEPARATION – A separation along bedding after extraction or exposure due to stress relief or slaking.

INCIPIENT JOINT (IJ) OR INCIPIENT FRACTURE (IF) – A joint or fracture which does not continue through the specimen or at least is not seen with the naked eye. However, when the specimen is wetted, and then allowed to dry, the joint or fracture trace is evident. When core is broken, it breaks along an existing plane.



RANDOM FRACTURE (RF) – A natural break which does not belong to a joint set, and which exhibits a generally rough, very irregular, nonplanar surface.

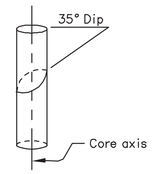


MECHANICAL BREAK (MB) – A break due to drilling, blasting, or handling. Mechanical breaks parallel to bedding or foliation are called Bedding Breaks (BB) or Foliation Breaks (FB), respectively. Recognizing mechanical breaks may be difficult. The absence of oxidation, staining, or mineral fillings, and often a hackly or irregular surface are clues for recognition.

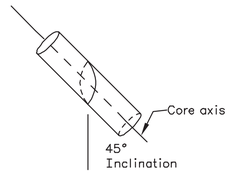


FRACTURE ZONE (FZ) – Numerous, very closely spaced intersecting fractures. Often fragmented core cannot be fitted together.

METHOD OF MEASURING DIP OF PLANAR DISCONTINUITIES, FOLIATION, AND BEDDING IN CORE



1. Vertical hole – true dip is measured and reported.

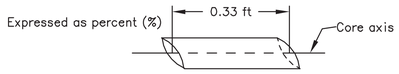


2. Angle hole – true dip usually not known; angle is measured from core axis and is called inclination.

ROCK QUALITY DESIGNATION (RQD)

EXAMPLE SHOWN FOR CORE, BUT APPLICABLE TO ANY LINEAR OBSERVATION

$$RQD = \frac{\text{Sum of length of solid core pieces} > 0.33 \text{ ft [4 in] (100 mm) long}}{\text{Length of the run in feet (mm)}} \times 100$$



FRACTURE FREQUENCY

FRACTURE FREQUENCY – The number of natural fractures occurring within a base length or core run. The number of fractures is divided by the length and is reported as fractures per foot or fractures per meter. Expressed as 3/m or 6/ft.

FRACTURE DENSITY

FRACTURE DENSITY – Based on the spacing of all natural fractures in an exposure or core recovery lengths in boreholes; excludes mechanical breaks, shears, and shear zones; however, shear-disturbed zones (fracturing outside the shear) are included. Descriptors for fracture density apply to all rock exposures such as tunnel walls, dozer trenches, outcrops, or foundation cut slopes and inverts, as well as boreholes. Descriptive criteria presented below are based on borehole cores where lengths are measured along the core axis. For other exposures the criterium is distance measured between fractures (size of blocks).

UNFRACTURED (FD0): No fractures.

VERY SLIGHTLY FRACTURED (FD1): Core recovered mostly in lengths greater than 3 feet (1 m).

SLIGHTLY TO VERY SLIGHTLY FRACTURED (FD2) *

SLIGHTLY FRACTURED (FD3): Core recovered mostly in lengths from 1 to 3 feet (300 to 1000 mm) with few scattered lengths less than 1 foot (300 mm) or greater than 3 feet (1000 mm).

MODERATELY TO SLIGHTLY FRACTURED (FD4) *

MODERATELY FRACTURED (FD5): Core recovered mostly in 0.3- to 1.0-foot (100- to 300-mm) lengths with most lengths about 0.6 foot (200 mm).

INTENSELY TO MODERATELY FRACTURED (FD6) *

INTENSELY FRACTURED (FD7): Lengths average from 0.1 to 0.3 foot (30 to 100 mm) with scattered fragmented intervals. Core recovered mostly in lengths less than 0.3 foot (100 mm).

VERY INTENSELY TO INTENSELY FRACTURED (FD8) *

VERY INTENSELY FRACTURED (FD9): Core recovered mostly as chips and fragments with a few scattered short core lengths.

---

* Combinations of fracture densities (e.g., Very Intensely to Intensely Fractured or Moderately to Slightly Fractured) are used where equal distribution of both fracture density characteristics are present over a significant interval or exposure, or where characteristics are "in between" the descriptor definitions.

FRACTURE SPACING

JOINT SET, OR FRACTURE SPACING DESCRIPTOR	TRUE SPACING
EXTREMELY WIDELY SPACED (SP1)	Greater then 10 ft (>3 m)
VERY WIDELY SPACED (SP2)	3 to 10 ft (1 to 3 m)
WIDELY SPACED (SP3)	1 to 3 ft (300 mm to 1 m)
MODERATELY SPACED (SP4)	0.3 to 1 m (100 to 300 mm)
CLOSELY SPACED (SP5)	0.1 to 0.3 ft (30 to 100 mm)
VERY CLOSELY SPACED (SP6)	less than 0.1 ft (<30 mm)

FRACTURE CONTINUITY

CONTINUITY DESCRIPTOR	DISCONTINUITY LENGTH
DISCONTINUOUS (C1)	Less than 3 ft (<1 m)
SLIGHTLY CONTINUOUS (C2)	3 to 10 ft (1 to 3 m)
MODERATELY CONTINUOUS (C3)	10 to 30 ft (3 to 10 m)
HIGHLY CONTINUOUS (C4)	30 to 100 ft (10 to 30 m)
VERY CONTINUOUS (C5)	Greater than 100 ft (>30 m)

FRACTURE ENDS (JOINT SURVEYS)

FRACTURE ENDS DESCRIPTOR	DESCRIPTIVE CRITERIA
EØ	Zero ends leave the exposure (both ends can be seen).
E1	One end of the fracture terminates in the exposure (one end can be seen).
E2	Neither fracture end terminates in the exposure (neither end can be seen).

FRACTURE OPENNESS OR FILLING THICKNESS

FILLING THICKNESS DESCRIPTOR	THICKNESS/OPENNESS	OPENNESS DESCRIPTOR
CLEAN (TØ)	No film or coating.	
	No visible separation.	TIGHT (OØ)
VERY THIN (T1)	Less than 0.003 ft [1/32 in] (<1 mm).	SLIGHTLY OPEN (O1)
MODERATELY THIN (T2)	0.003 to 0.01 ft [1/32 to 1/8 in] (1 to 3 mm).	MODERATELY OPEN (O2)
THIN (T3)	0.01 to 0.03 ft [1/8 to 3/8 in] (3 to 10 mm).	OPEN (O3)
MODERATELY THICK (T4)	0.03 ft [3/8 in] to 0.1 ft (10 to 30 mm).	MODERATELY WIDE (O4)
THICK (T5)	Greater than 0.1 ft (>30 mm). Actual thickness or openings recorded.	WIDE (O5)

FRACTURE MOISTURE CONDITIONS

MOISTURE DESCRIPTOR	DESCRIPTIVE CRITERIA
M1	The fracture is dry. It is tight or filling (where present) is of sufficient density or composition to impede waterflow. Waterflow along the fracture does not appear possible.
M2	The fracture is dry with no evidence of previous waterflow. Waterflow appears possible.
M3	The fracture is dry, but shows evidence of waterflow such as staining, leaching and/or vegetation.
M4	The fracture or filling (where present) is damp, but no free water is present.
M5	The fracture shows seepage. It is wet with occasional drops of water.
M6	The fracture emits a continuous flow (estimate flow rate) under low pressure. Filling materials (where present) may show signs of leaching or piping.
M7	The fracture emits a continuous flow (estimate flow rate) under moderate to high pressure. Water is squirting and/or filling material (where present) may be substantially washed out.

FRACTURE ROUGHNESS

Refers to small-scale asperities of surfaces, not large-scale undulations or waviness.

STEPPED (R1): Near-normal steps and ridges occur on the fracture surface.
ROUGH (R2): Large, angular asperities can be seen.
MODERATELY ROUGH (R3): Asperities are clearly visible and fracture surface feels abrasive.
SLIGHTLY ROUGH (R4): Small asperities on the fracture surface are visible and can be felt.
SMOOTH (R5): No asperities, smooth to the touch.
POLISHED (R6): Extremely smooth and shiny.

FRACTURE SURFACE AND/OR FILLING ALTERATION AND HARDNESS

Descriptors for weathering or alteration of fracture surfaces and fracture fillings (excluding soil materials) are the same as those used for weathering and alteration of rock.

Descriptors for hardness/strength of fillings and/or fracture surfaces are the same as those presented for hardness of rock and consistency of soils.

DISCONTINUITY HEALING

TOTALLY HEALED (HL1) – All fragments bonded, discontinuity is completely healed or recemented to a degree at least as hard as surrounding rock.

MODERATELY HEALED (HL3) – Greater than 50 percent of fractured or sheared material, discontinuity surfaces or filling is healed or recemented; and/or strength of healing agent is less hard than surrounding rock.

PARTLY HEALED (HL5) – Less than 50 percent of fractured or sheared material, discontinuity surface or filling is healed or recemented.

NOT HEALED (HL6) – Discontinuity surface, fractured zone, sheared material or filling is not healed or recemented, rock fragments or filling (if present) held in place by their own angularity and/or cohesiveness.

SHEAR/FAULT DESCRIPTORS

SHEAR/FAULT GOUGE CONSISTENCY

DESCRIPTOR	DESCRIPTIVE CRITERIA (Similar to consistency of soils)
VERY HARD	Gouge cannot be broken with finger pressure; cannot be indented with fingernail.
HARD	Gouge can be broken with firm finger pressure; can be indented with fingernail; cannot be indented with thumb.
FIRM	Gouge can be easily crumbled; can be indented with thumb 1 to 5 mm.
SOFT	Gouge can be easily molded; can be penetrated with thumb 5 to 25 mm.
VERY SOFT	Gouge can be penetrated with thumb more than 25 mm.

SHEAR/FAULT MOISTURE DESCRIPTORS

The apparent moisture content of gouge is described as WET (visible free water); MOIST (damp, but no visible water); and DRY (absence of moisture, dusty, dry to the touch). Moisture descriptors M1 through M7 may be used to describe the shear or shear zone.

BRECCIA SHAPES

Angular . . . . . 

Subangular . . . . . 

Subrounded . . . . . 

Rounded . . . . . 

Platy . . . . . 

Lens-shaped . . . . . 

Wedge-shaped . . . . . 

Contorted . . . . . 

REV NO		
1	3-8-00 D – P. M. R.	CONVERTED ORIGINAL DRAWING 40-D-6499 TO ACAD, CHANGED DWG. NO., MINOR REVISIONS.

ALWAYS THINK SAFETY

UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

GEOLOGY FOR DESIGN & SPECIFICATIONS

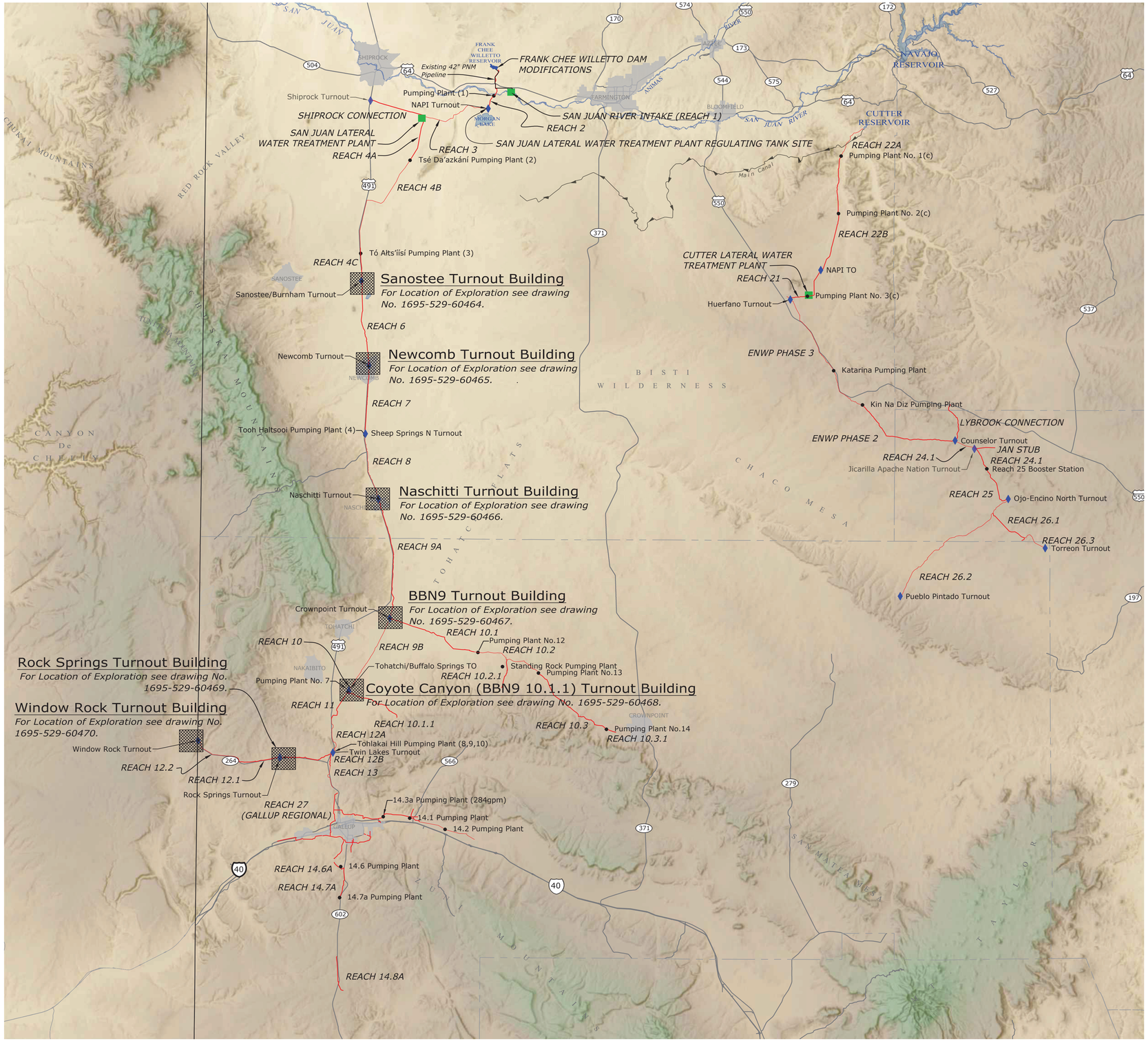
STANDARD DESCRIPTORS AND DESCRIPTIVE CRITERIA FOR DISCONTINUITIES

GEOLOGY NOMENCLATURE COMMITTEE _ CHECKED_ CHUCK SULLIVAN

DRAWN _MARSHALL MONSON_ TECH. APPROVAL _PETER M. ROHRER

APPROVED_ _MARK McKEOWN_ (PEER REVIEWER)

CADD SYSTEM	CADD FILENAME	DATE AND TIME PLOTTED
AutoCAD Rel. 15.0	40-D-7023.DWG	JUNE 26, 2002 11:10
DENVER, COLORADO	MARCH 8, 2000	40-D-7023





KEY MAP



LEGEND

- Pipeline (Varied by Reach)
- Water Treatment Plant
- Pumping Plant Location
- Turnout Location
- Highway (as designated)
- County Line

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

D. Thornburg	DESIGNED
D. Thornburg	DRAWN
Beyer, Christopher	CHECKED
Beyer, Christopher R	TECH. APPR. Geologist
Gladden, Ryan L	APPROVED P.E. - Professional Engineer ADMIN APPROVAL

FARMINGTON, NM

GENERAL VICINITY MAP

1695-529-60459

SHEET 1

LAST SAVED DATE
2025-09-04
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AUTOCAD CIVIL 3D (LMS TECH)
CAD FILE NAME
1695-529-60459-RD-1.DWG

GENERAL GEOLOGIC LEGEND

STRATIGRAPHY

Qal **Quaternary Alluvium**

Quaternary alluvium is derived from a variety of sources including slope wash, eolian and colluvial deposition as well as weathering and decomposition of in-place bedrock. Alluvium contains variable amounts of plastic to nonplastic fines and may contain sand, gravel, cobbles, and boulders.

Kmf **Cretaceous Menefee Formation**

The Cretaceous Menefee Formation consists primarily of light brown, fine-grained sandstone with interbedded claystone, shale and siltstone but can vary in color and grain size. The sandstone is generally moderately weathered (W5),moderately soft (H5) to soft (H6) and slightly fractured at the surface. The same sandstone unit, when excavated below ground level, can occasionally be moderately hard (H4) and slightly weathered (W3). A hard, calcareous sandstone bed frequently caps the upper 1 to 3 feet in some areas and is occasionally interbedded or lensed within the softer sandstone units. Intermittent hard (H3) concretionary lenses are are also present throughout the formation. This caprock/interbedded sandstone is often hard (H3) and can only be broken by a heavy hammer blow. The claystone is sandy, light brown to grey in color and is generally laminated to thinly bedded, intensely to moderately weathered (W6) and moderately soft (H5) to soft (H6). The claystone often grades to siltstone and occasionally becomes fissile, grading to shale. Thin coal seams are also sometimes encountered. The coal is subbituminous to bituminous, friable, soft (H6), and moderately to intensely weathered (W6) to moderately to slightly weathered (W4).

Fill **Backfill material from pipeline construction**

Fill consists of varying native materials including intermixed, lensed, clay, silt, sand, gravel, and cobble material. Large particles are primarily subangular to angular, moderately hard (H4) to hard (H3) sandstone. Sporadic construction debris including garbage and concrete fragments are sometimes encountered.

GEOLOGIC EXPLANATION

TPTB-25-5 ■ — *Test Pit location with Identification number*

Qal / Kmf — *Geologic Unit Contact*

— Qal / Kmf — *Dashed where approximate*

— ? Qal / Kmf ? — ? — ? — *Queried where inferred*

GENERAL GEOLOGIC NOTES

Geologic standards and definitions used for rock quality and rock discontinuities are based on Bureau of Reclamation Engineering Geology Field Manual and drawing numbers 40-D-6493 and 40-D-6499.

Soil classification and descriptions are based on the Unified Soil Classification System (USCS) and Bureau of Reclamation procedures and guidelines as described in Geotechnical Branch Training Manuals Nos. 4, 5, 6, and Designations USBR 5000 (laboratory classification) and USBR 5005 (visual classification). These procedures are similar to ASTM D2487-11 and ASTM D2488-09A respectively.

Interpretations shown on geologic profiles are based on Test Pit, Drill Hole, outcrop, and field mapping data. Interpretations suggest general trends between data points and do not depict localized irregularities.

For the entire description of material, drilling or excavation method and conditions, exact locations of the hole, etc., see complete log.

BEDROCK ABBREVIATIONS:

SS- SANDSTONE
CLSTN - CLAYSTONE
SLTST - SILTSTONE

— BUREAU OF — RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO		
2	2025-09-23 D. Thornburg	529	UPDATED THE FILL DESCRIPTION IN THE LEGEND
1	2025-09-22	529	CORRECTED TYPOGRAPHICAL ERRORS IN THE LEGEND AND NOTES

Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Graham, Kris
CHECKED
Beyer, Christopher R
TECH. APPR.
Gladden, Ryan L P.E.
APPROVED
ADMIN APPROVAL

FARMINGTON, NM 2025-09-22

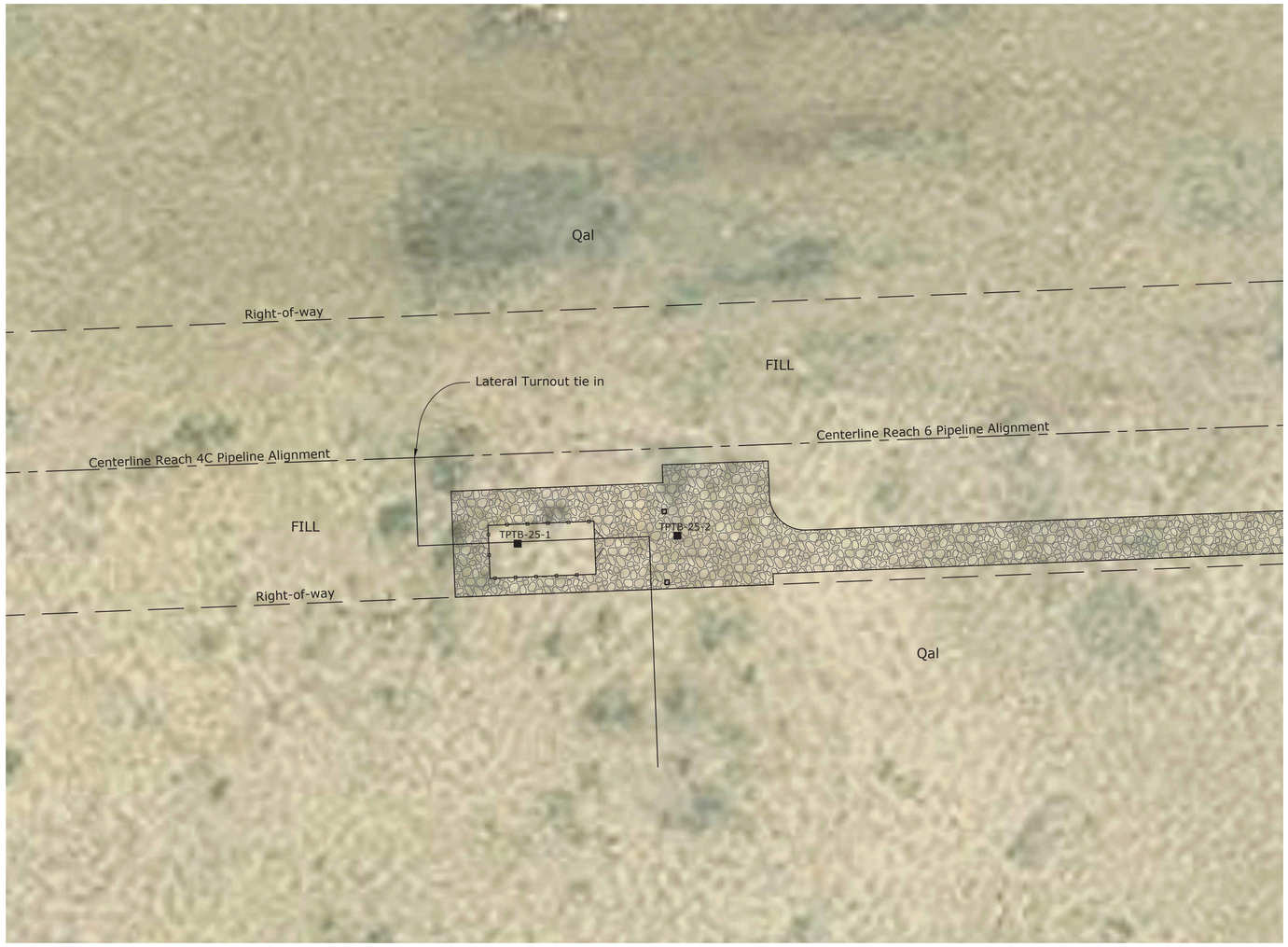
GENERAL GEOLOGIC LEGEND, EXPLANATION, AND NOTES

1695-529-60463

SHEET 1

LAST SAVED DATE
2025-09-23
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-529-60463-RD-1.DWG



SANOSTEE TURNOUT BUILDING SITE - PLAN VIEW

TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-1	1976766.41	2462192.47	W108° 43' 12.74"	N36° 25' 45.47"	W108.72020422	N36.42929585
	TPTB-25-2	1976721.19	2462194.72	W108° 43' 12.70"	N36° 25' 45.02"	W108.72019516	N36.42917172

0 20 40 Feet

GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999713762328.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

BUREAU OF RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL
FARMINGTON, NM

SANOSTEE TURNOUT

SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60464

SHEET 1

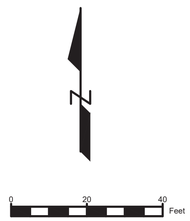
LAST SAVED DATE
2023-05-04
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-529-60464-RD-1.DWG



NEWCOMB TURNOUT BUILDING SITE - PLAN VIEW

TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-3	1925478.39	2466227.26	W108° 42' 17.70"	N36° 17' 18.68"	W108.70491804	N36.28852361
	TPTB-25-4	1925429.92	2466228.34	W108° 42' 17.69"	N36° 17' 18.21"	W108.70491288	N36.28839050



GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999729101996.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

— BUREAU OF — RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL
FARMINGTON, NM

NEWCOMB TURNOUT

SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60465

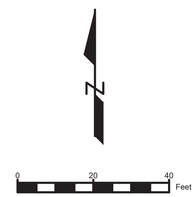
SHEET 1

LAST SAVED DATE
2019-10-04
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AUTOCAD CIVIL 3DS (LMS TECH)
CAD FILENAME
1695-529-60465-RD-1.DWG



NASCHITTI TURNOUT BUILDING SITE - PLAN VIEW



TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-5	1845403.30	2472011.78	W108° 40' 58.49"	N36° 04' 07.37"	W108.68291393	N36.06871406
	TPTB-25-6	1845336.50	2472034.89	W108° 40' 58.20"	N36° 04' 06.71"	W108.68283377	N36.06853112

GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999710324996.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

BUREAU OF RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL
FARMINGTON, NM

NASCHITTI TURNOUT

SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60466

SHEET 1

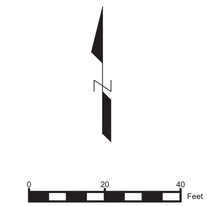
LAST SAVED DATE
2025-09-04
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AUTOCAD CIVIL (LMS TECH)
CAD FILENAME
1695-529-60466-RD-1.DWG



BBN9 TURNOUT BUILDING SITE - PLAN VIEW

TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-7	1772888.78	2479029.36	W108° 39' 25.56"	N35° 52' 10.88"	W108.65709877	N35.86968816



GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999703693004.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

— BUREAU OF — RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL

FARMINGTON, NM

BBN9 TURNOUT

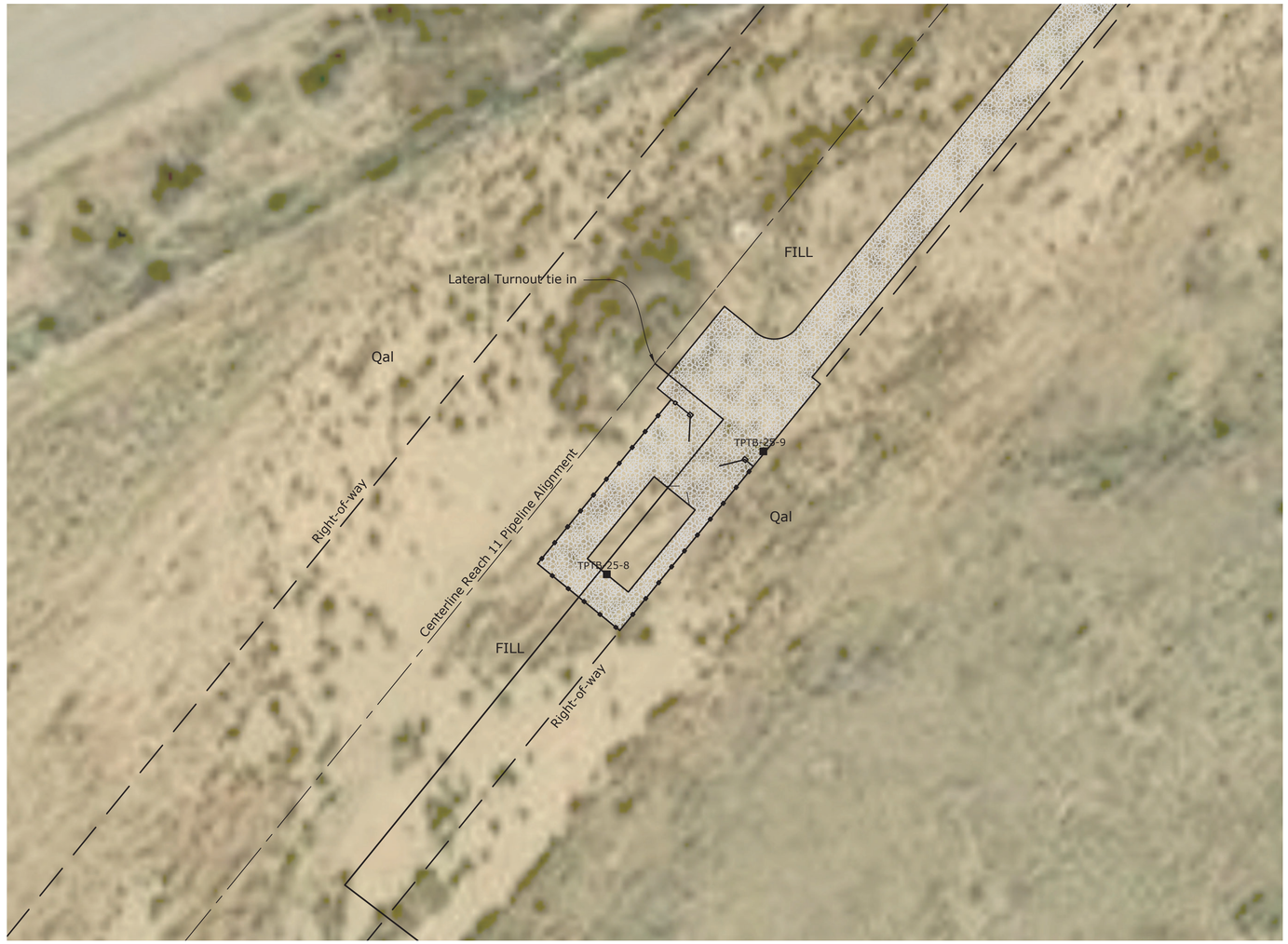
SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60467

SHEET 1

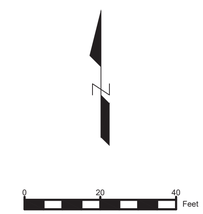
LAST SAVED DATE
7/31/2024 9:04 AM
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AUTOCAD LDS (LMS TECH)
CAD FILENAME
1695-529-60467-RD-1.DWG



COYOTE CANYON (BBN9) TURNOUT BUILDING SITE - PLAN VIEW

TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-8	1728102.86	2453520.00	W108° 44' 30.43"	N35° 44' 45.75"	W108.74178712	N35.74604287
	TPTB-25-9	1728137.30	2453564.68	W108° 44' 29.90"	N35° 44' 46.10"	W108.74163764	N35.74613863



GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999703693004.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

BUREAU OF RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL
FARMINGTON, NM 2025-07-07

COYOTE CANYON (BBN9) TURNOUT

SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60468

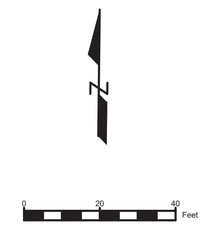
SHEET 1

LAST SAVED DATE
2025-09-04
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-529-60468-RD-1.DWG



ROCK SPRINGS TURNOUT BUILDING SITE - PLAN VIEW



TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-10	1688615.00	2415232.13	W108° 52' 09.82"	N35° 38' 11.51"	W108.86939527	N35.63653078

GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999699931998.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

BUREAU OF RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L.
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL
FARMINGTON, NM 2025-07-07

ROCK SPRINGS TURNOUT

SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60469

SHEET 1

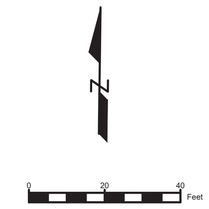
LAST SAVED DATE
2025-07-07 09:14
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-529-60469-RD-1.DWG



WINDOW ROCK TURNOUT BUILDING SITE - PLAN VIEW

TEST PIT LOCATIONS							
	Description	*Northing*	*Easting*	*Longitude*	*Latitude*	*Longitude (DEC DEG)*	*Latitude (DEC DEG)*
	TPTB-25-11	1698982.19	2363039.75	W109° 02' 43.58"	N35° 39' 48.13"	W109.04543962	N35.66336813
	TPTB-25-12	1698971.85	2363075.44	W109° 02' 43.15"	N35° 39' 48.03"	W109.04531904	N35.66334094



GENERAL NOTES

1. *Coordinates based on New Mexico State Plane West Zone, feet. (NAD83).*
2. *The Ground to New Mexico State Plane West Zone Grid conversion factor is 0.999720158003.*

REFERENCE DRAWINGS

General Location Map.....1695-529-60459.

General Geologic Legend, Explanation, and Notes.....1695-529-60463.

— BUREAU OF — RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Graham, Kris / Beyer, Christopher
GEOLOGY
D. Thornburg
DRAWN
Beyer, Christopher
CHECKED
Beyer, Christopher R
TECH. APPR. Geologist
Gladden, Ryan L
APPROVED P.E. - Professional Engineer
ADMIN APPROVAL

FARMINGTON, NM 2025-07-07

WINDOW ROCK TURNOUT

SURFACE GEOLOGY & LOCATION OF EXPLORATION

1695-529-60470

SHEET 1

LAST SAVED DATE
2025-06-04
LAST SAVED BY
DTHORNBURG

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-529-60470-RD-1.DWG

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-1

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Sonostee Turnout
GROUND ELEVATION: 5564.7 ft
COORDINATES: N 1,976,766.4 E 2,462,192.5
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X8.5'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/31/2025 DATE: NE
DATE EXCAVATED: 7/31/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1	(CL)s	0.0 to 2.2 ft LEAN CLAY WITH SAND: About 75% fines; with medium plasticity, medium dry strength, and medium toughness; about 25% fine sand; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) /BACKFILL.			
2	2.2 ft ()				
3	SP-SM	2.2 to 5.5 ft POORLY GRADED SAND WITH SILT: About 90% fine sand; about 10% nonplastic fines; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.5 ft. Total: 91.6 lb/ft³, 2.6% (83.3% compaction) LAB TEST DATA: 65.3% sand, 34.7% fines, 0 % gravel, LL = NP PI =NP SPG = 2.62 Maximum dry density: 109.9 lb/ft³, optimum water content = 13.6% Laboratory classification is SILTY SAND.			
4	Corrosivity sample and in-place density test at 4.5 ft.				
5	5.5 ft ()	GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / BACKFILL.			
6	SS	5.5 to 8.5 ft SANDSTONE: Fine grained, thinly bedded with claystone interbeds, argillaceous, light brown in color, soft (H6) to moderately soft (H5) and intensely weathered (W7) to moderately to intensely weathered (W6), strong reaction with HCl, recovered as 3- to-6 inch subangular fragments. GEOLOGIC INTERPRETATION: Cretaceous Menefee Formation (Kmf).			
7					
8	8.5 ft ()				

COMMENTS: Surface scarcely vegetated with weeds. Discontinued excavation due to difficult excavation in sandstone bedrock.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-2

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Sonostee Turnout
GROUND ELEVATION: 5564.8 ft
COORDINATES: N 1,976,721.2 E 2,462,194.7
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X9'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/31/2025 DATE: NE
DATE EXCAVATED: 7/31/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1	(CL)s	0.0 to 3.3 ft LEAN CLAY WITH SAND: About 75% fines with medium plasticity, medium dry strength, and medium toughness; about 25% fine sand; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) /BACKFILL.			
2					
3	3.3 ft ()				
4	SP-SM Corrosivity sample and in-place density test at 4.0 ft.	ft POORLY GRADED SAND WITH SILT: About 90% fine sand; about 10% nonplastic fines; maximum size, fine sand; Strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.0 ft. Total: 86.3 lb/ft³, 3.7 % (73.8 % compaction) LAB TEST DATA: 60.0% sand, 40.0 % fines, 0 % gravel, LL=NP PI = NP, SPG=2.63 Maximum dry density: 116.9 lb/ft³, optimum water content = 13.3 % Laboratory classification is SILTY SAND.			
5					
6	6.0 ft ()				
7	(CL)s	GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / BACKFILL. 6.0 to 9.0 ft LEAN CLAY WITH SAND: About 75% fines; with medium plasticity, medium dry strength, and medium toughness; about 25% fine sand; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) /BACKFILL.			
8					
9	9.0 ft ()				

COMMENTS: Surface scarcely vegetated with weeds. Discontinued excavation due to difficult excavation in very stiff dry clay.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-3

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Newcomb Turnout
GROUND ELEVATION: 5563.5 ft
COORDINATES: N 1,925,478.4 E 2,466,227.3
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X11'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/31/2025 DATE: NE
DATE EXCAVATED: 7/31/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1–8	SC	0.0 to 8.5 ft CLAYEY SAND: About 80% fine sand; about 20% fines with low plasticity, low dry strength, and low toughness; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, homogeneous, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 5.0 ft. Total: 90.4 lb/ft³, 6.3 % (79.0% compaction) LAB TEST DATA: 54.6 % sand, 45.4 % fines, 0 % gravel, LL= 26.6 PI = 11.5 SPG=2.60 Maximum dry density: 114.4 lb/ft³, optimum water content = 15.0 % Laboratory classification is CLAYEY SAND. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
4	Corrosivity sample at 4.0 ft. In-place density test at 5.0 ft.				
8	8.5 ft ()				
9	SP	8.5 to 9.5 ft POORLY GRADED SAND: About 95% fine sand; about 5% nonplastic fines; trace medium to coarse sand and gravel; maximum size, 25mm, strong reaction with HCl. IN-PLACE CONDITION: Light brown in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
9	9.5 ft ()				
10	SC	9.5 to 11.0 ft CLAYEY SAND: About 80% fine sand; about 20% fines with low plasticity, low dry strength, and low toughness; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
11	11.0 ft ()				

COMMENTS: Surface free of vegetation. Test pit in dirt road. Discontinued excavation due to limit of equipment.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-4

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Newcomb Turnout
GROUND ELEVATION: 5563.4 ft
COORDINATES: N 1,925,429.9 E 2,466,228.3
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X11.5'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/31/2025 DATE: NE
DATE EXCAVATED: 7/31/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1	SC	0.0 to 3.0 ft CLAYEY SAND: About 80% fine sand; about 20% fines with low plasticity, low dry strength, and low toughness; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
2					
3	3.0 ft ()				
4	SP-SM	3.0 to 11.5 ft POORLY GRADED SAND WITH SILT: About 90% fine sand; about 10% fines with no plasticity, no dry strength and rapid dilatancy; trace medium to coarse sand and gravel; maximum size, 25mm, strong reaction with HCl. IN-PLACE CONDITION: Light brown in color, Lensed with uneven bedding, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.0 ft. Total: 91.0 lb/ft³, 6.9 % (79.1% compaction) LAB TEST DATA: 57.7 % sand, 42.3% fines, 0 % gravel, LL= NP PI = NP SPG=2.63 Maximum dry density: 115.1 lb/ft³, optimum water content = 14.1 % Laboratory classification is SILTY SAND. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
5					
6					
7	Corrosivity sample and in-place density test at 4.0 ft.				
8					
9					
10					
11	11.5 ft ()				

COMMENTS: Surface free of vegetation. Test pit in dirt road. Discontinued excavation due to limit of equipment. Encountered 3" pvc pipe painted orange around 6 ft in depth. Pipe was empty and did not appear to continue on other side of pit. Likely garbage.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-5

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Naschitti Turnout
GROUND ELEVATION: 5872.4 ft
COORDINATES: N 1,845,403.3 E 2,472,011.8
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X10'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/30/2025 DATE: NE
DATE EXCAVATED: 7/30/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1	SC	0.0 to 1.6 ft CLAYEY SAND: About 80% fine sand; about 20% fines with low plasticity, medium dry strength, and low toughness; trace gravel and cobbles; maximum size, 150mm; strong reaction with HCl. IN-PLACE CONDITION: Brownish gray in color, homogeneous, dry.	Trace	Trace	
1.6 ft ()					
2	SM	GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / BACKFILL. 1.6 to 9.7 ft SILTY SAND: About 80% fine sand; about 20% nonplastic fines; trace gravel and cobbles; maximum size, 150mm; strong reaction with HCl. IN-PLACE CONDITION: Brownish gray in color, homogeneous, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 5.0 ft. Total: 93.8 lb/ft³, 6.4 % (87.2 % compaction) LAB TEST DATA: 74.3 % sand, 25.7 % fines, 0 % gravel, LL= NP, PI = NP SPG=2.61 Maximum dry density: 107.6 lb/ft³, optimum water content = 13.5 % Laboratory classification is SILTY SAND. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal)/ BACKFILL.	Trace		
3					
4					
5	Corrosivity sample at 4.5 ft. In-place density test at 5.0 ft.				
6					
7					
8					
9					
9.7 ft ()					
10 ft ()	SS	9.7 to 10.0 ft SANDSTONE: Fine grained, thinly bedded, argillaceous, light brown in color, soft (H6) to moderately soft (H5) and intensely weathered (W7) to moderately to intensely weathered (W6), strong reaction with HCl, recovered as 3- to -6 inch subangular fragments. GEOLOGIC INTERPRETATION: Cretaceous Menefee Formation (Kmf).			
10					

COMMENTS: Surface scarcely vegetated with weeds. Some gravel, cobbles and concrete fragments on the surface. Discontinued excavation due to refusal in sandstone bedrock.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-6

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Naschitti Turnout
GROUND ELEVATION: 5873.3 ft
COORDINATES: N 1,845,336.5 E 2,472,034.9
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X6.5'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/30/2025 DATE: NE
DATE EXCAVATED: 7/30/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1	SC 1.5 ft ()	0.0 to 1.5 ft CLAYEY SAND: About 80% fine sand; about 20% fines with low plasticity, medium dry strength, and low toughness; trace gravel and cobbles; maximum size, 150mm; strong reaction with HCl. IN-PLACE CONDITION: Brownish gray in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / BACKFILL.	Trace	Trace	
2 3 4	SM Corrosivity sample at 4.0 ft. in-place density test at 4.5 ft. 4.5 ft ()	1.5 to 4.5 ft SILTY SAND: About 80% fine sand; about 20% nonplastic fines; trace gravel and cobbles; maximum size, 150mm; strong reaction with HCl. IN-PLACE CONDITION: Brownish gray in color, homogeneous, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.5 ft. Total: 99.4 lb/ft³, 6.0 % (88.6 % compaction) LAB TEST DATA: 82.7 % sand, 17.3 % fines, 0 % gravel, LL= NP, PI = NP, SPG= 2.62. Maximum dry density: 112.2 lb/ft³, optimum water content =15.5 %. Laboratory classification is SILTY SAND. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal)/ BACKFILL.	Trace		
5	SC 5.5 ft ()	4.5 to 5.5 ft CLAYEY SAND: About 70% fine sand; about 30% fines with low plasticity, medium dry strength, and low toughness; trace gravel and cobbles; maximum size, 150mm; strong reaction with HCl. IN-PLACE CONDITION: Brownish gray in color, homogeneous, dry.	Trace		
6	SS 6.5 ft ()	GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / BACKFILL. 5.5 to 6.5 ft SANDSTONE: Fine grained, thinly to moderately bedded, light brown in color, soft (H6) to moderately soft (H5) and intensely weathered (W7) to moderately weathered (W5), strong reaction with HCl, recovered as 3- to -6 inch subangular fragments. GEOLOGIC INTERPRETATION: Cretaceous Menefee Formation (Kmf).			

COMMENTS: Surface scarcely vegetated with weeds. Some gravel, cobbles and concrete fragments on the surface. Discontinued excavation due to refusal in sandstone bedrock.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-7

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: BBN9 Turnout
GROUND ELEVATION: 6049.5 ft
COORDINATES: N 1,772,888.8 E 2,479,029.4
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X12'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/30/2025 DATE: NE
DATE EXCAVATED: 7/30/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1–8	SC Corrosivity sample at 4.0 ft. In-place density test at 4.5 ft. 8.5 ft ()	0.0 t 8.5 ft CLAYEY SAND: About 85% fine sand; about 15% fines with low plasticity, low dry strength, and low toughness; maximum size, fine sand; strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry from 0.0 to 4.0 ft. Moist from 4.0 to 8.5 ft. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.5 ft. Total: 84.1 lb/ft³, 14.8 % (75.9 % compaction) LAB TEST DATA: 43.0 % sand, 57.0 % fines, 0 % gravel, LL= 32.1 PI = 15.9 SPG= 2.67. Maximum dry density: 110.8 lb/ft³, optimum water content = 16.0 %. Laboratory classification is SANDY LEAN CLAY . GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
9–12	SP-SM 12.0 ft ()	8.5 to 12.0 ft POORLY GRADED SAND WITH SILT: About 90% fine sand; about 10% nonplastic fines; maximum size, fine sand; Strong reaction with HCl. IN-PLACE CONDITION: Gray in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / BACKFILL.			

COMMENTS: Surface vegetated with Russian Thistle and native plants. Discontinued excavation due to limit of equipment.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-8

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Coyote Canyon Turnout
GROUND ELEVATION: 6251.4 ft
COORDINATES: N 1,728,102.9 E 2,453,520
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X10'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/29/2025 DATE: NE
DATE EXCAVATED: 7/29/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
	SC	0.0 to 0.8 ft CLAYEY SAND: About 85% fine sand; about 15% fines with low plasticity, low dry strength, and low toughness; maximum size, fine sand; strong reaction with HCl.			
	0.8 ft ()				
CEMENT/GRAVEL/SOIL		IN-PLACE CONDITION: Gray in color, homogeneous, dry.			
	1.5 ft ()				
	SC	GEOLOGIC INTERPRETATION: BACKFILL.			
2		0.8 to 1.5 ft CONCRETE/CEMENT TRUCK WASHOUT: Cement and gravel mixed with native soil. Evidence that concrete truck was washed out at test pit site then covered up with soil. moderately soft (H5), fractured and discontinuous.			
3		IN-PLACE CONDITION: Gray in color, moderately cemented, dry.			
		GEOLOGIC INTERPRETATION: Concrete Truck washout.			
4		1.5 to 10.0 ft CLAYEY SAND: About 85% fine sand; about 15% fines with low plasticity, low toughness and low dry strength and slow dilatancy; maximum size, fine sand; strong reaction with HCl.			
5		IN-PLACE CONDITION: Brownish red in color, homogeneous, dry.			
	Corrosivity sample and in-place density test at at 4.5 ft.	IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.5 ft. Total: 92.8 lb/ft³, 6.3 % (82.5 % compaction) LAB TEST DATA: 35.3% sand, 64.7 % fines, 0 % gravel, LL=27.9 PI = 10.6 SPG=2.66 Maximum dry density: 112.5 lb/ft³, optimum water content = 14.4 % Laboratory classification is SANDY LEAN CLAY.			
6					
7		GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal)/ BACKFILL.			
8					
9					
	10.0 ft ()				
10					

COMMENTS: Surface sparsely vegetated with Russian Thistle. Gravel and concrete fragments at the surface.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-9

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
LOCATION: Coyote Canyon Turnout
COORDINATES: N 1,728,137.3 E 2,453,564.7
APPROXIMATE DIMENSIONS: 12'X10'X11.5'
DEPTH TO WATER: 7/30/2025 DATE: NE

PROJECT: Navajo Gallup Water Supply Project
GROUND ELEVATION: 6249.5 ft
METHOD OF EXPLORATION: Deere 310G Backhoe
LOGGED BY: C. Beyer
DATE EXCAVATED: 7/29/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
1	SC 1.0 ft ()	0.0 to 1.0 ft CLAYEY SAND: About 85% predominately fine sand, trace medium to coarse sand, gravel and cobbles; about 15% fines with low plasticity, low dry strength, and low toughness; maximum size, 200mm; strong reaction with HCl.	Trace	Trace	
2	(SM)g	IN-PLACE CONDITION: Brownish red in color, dry. GEOLOGIC INTERPRETATION: BACKFILL.			
3	3.5 ft ()	1.0 to 3.5 ft SILTY SAND WITH GRAVEL: About 65% fine sand; about 20 % fine to coarse gravel; about 15% nonplastic fines; maximum size, coarse gravel; strong reaction with HCl.			
4	SM	IN-PLACE CONDITION: Brownish red in color, lensed, dry. GEOLOGIC INTERPRETATION: BACKFILL.			
5		3.5 to 11.5 ft SILTY SAND: About 85% fine sand; about 15% nonplastic fines; trace gravel and concrete fragments; maximum size, 75mm; strong reaction with HCl.			
6		IN-PLACE CONDITION: Brownish red in color, calcite cemented, lensed, dry.			
7	Corrosivity sample at 4.0 ft. In-place density test at at 4.5 ft.	IN-PLACE UNIT WEIGHT AND MOISTURE FROM 4.5 ft. Total: 94.3 lb/ft³, 6.3 % (80.6 % compaction) LAB TEST DATA: 54.5 % sand, 35.1 % fines, 10.4 % gravel, LL=21.6 PI =4.8 SPG =2.55. Maximum dry density: 117.0 lb/ft³, optimum water content = 11.0 %. Laboratory classification is CLAYEY SAND WITH SILT.			
8		GEOLOGIC INTERPRETATION: BACKFILL.			
9					
10					
11	11.5 ft ()				

COMMENTS: Surface sparsely vegetated with Russian Thistle. Gravel and concrete fragments at the surface.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-10

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Rock Springs Turnout
GROUND ELEVATION: 6852.9 ft
COORDINATES: N 1,688,615 E 2,415,232.1
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X10'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/28/2025 DATE: NE
DATE EXCAVATED: 7/28/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
0	SC	0.0 to 2.0 ft CLAYEY SAND: About 80% predominately fine sand; trace medium to coarse sand, gravel and cobbles; about 20% fines with low plasticity, low dry strength, and low toughness; maximum size, 250mm; strong reaction with HCl. IN-PLACE CONDITION: Brown in color, dry.	trace	trace	
2.0 ft ()		GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
2	(SC)g	2.0 to 10.0 ft CLAYEY SAND WITH GRAVEL: About 55% fine to coarse sand; about 35% fines with medium plasticity, medium dry strength, and medium toughness; maximum size, coarse gravel; asbout 10% fine to coarse gravel; maximum size coarse gravel; strong reaction with HCl. IN-PLACE CONDITION: Brown in color, lenses with gravel, moist. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 5.5 ft. Total: 92.6 lb/ft³, 2.4% (84.3% compaction) LAB TEST DATA: 36.9% sand, 63.1% fines, 0% gravel, LL=34.4 PI = 16.6 SPG=2.65 Maximum dry density: 109.8 lb/ft³, optimum water content = 12.6 % Laboratory classification is SANDY LEAN CLAY . GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
3					
4					
5					
6	Corrosivity sample at 4.0 ft. In-place density test at 5.5 ft.				
7					
8					
9					
10					
10.0 ft ()					

COMMENTS: Surface scarcely vegetated with weeds. Discontinued excavation due to tough conditions and backhoe overheating. Yellow cathodic protection tape discovered in first attempt at 4.0 feet in depth. Moved test pit about 10 feet east and started over.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-11

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Window Rock Turnout
GROUND ELEVATION: 6787.0 ft
COORDINATES: N 1,698,982.2 E 2,363,039.7
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X12'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/28/2025 DATE: NE
DATE EXCAVATED: 7/28/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
0	SP	0.0 to 4.0 ft POORLY GRADED SAND: About 95% fine sand; about 5% nonplastic fines; maximum size, fine sand, no to strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal) / FILL.			
	Corrosivity sample at 4.0 ft				
4.0 ft	()				
4.5 ft	()	4.0 to 4.5 ft POORLY GRADED SAND WITH GRAVEL: About 65% fine sand; About 30% fine to coarse subangular to subrounded gravel; About 5% fines with no plasticity, no dry strength and rapid dilatancy; trace cobbles; maximum size, 150mm, strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal).	trace	trace	
	(SP)g SP	4.5 to 12.0 ft POORLY GRADED SAND: About 90% fine sand; about 10% nonplastic fines; maximum size, fine sand, strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, homogeneous, dry. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal).			
12.0 ft	()				

COMMENTS: Surface scarcely vegetated with Russian Thistle. Discontinued excavation due to limit of equipment.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

7-1336-A (1-86)
Bureau of Reclamation

LOG OF TEST PIT NO. TPTB-25-12

SHEET 1 OF 1

FEATURE: San Juan lateral Turnout Buildings
PROJECT: Navajo Gallup Water Supply Project
LOCATION: Window Rock Turnout
GROUND ELEVATION: 6787.0 ft
COORDINATES: N 1,698,971.9 E 2,363,075.4
METHOD OF EXPLORATION: Deere 310G Backhoe
APPROXIMATE DIMENSIONS: 12'X10'X12'
LOGGED BY: C. Beyer
DEPTH TO WATER: 7/29/2025 DATE: NE
DATE EXCAVATED: 7/29/2025

DEPTH	CLASSIFICATION GROUP SYMBOL	CLASSIFICATION AND DESCRIPTION OF MATERIAL (SEE USBR 5000, 5005)	% PLUS 3 in (BY VOLUME) 3 - 5 in	5 - 12 in	PLUS 12 in
0	SP-SM	0.0 to 12.0 ft POORLY GRADED SAND WITH SILT : About 90% fine sand; about 10% nonplastic fines; trace fine to coarse gravel; maximum size, 75mm, no to strong reaction with HCl. IN-PLACE CONDITION: Brownish red in color, homogeneous, dry. IN-PLACE UNIT WEIGHT AND MOISTURE FROM 7.0 ft. Total: 92.3 lb/ft³, 2.4 % (80.7% compaction) LAB TEST DATA: 79.9 % sand, 20.1 % fines, 0% gravel, LL=NP PI =NP SPG=2.65 Maximum dry density: 114.3 lb/ft³, optimum water content = 10.8 % Laboratory classification is SILTY SAND. GEOLOGIC INTERPRETATION: Quaternary Alluvium (Qal).			
6	Corrosivity sample at 4.0 ft. In-place density test at at 7.0 ft.				
12	12.0 ft ()				

COMMENTS: Surface scarcely vegetated with Russian Thistle. Discontinued excavation due to limit of equipment.

BUREAU OF RECLAMATION

TEST PIT SAN JUAN LATERAL TURNOUT BUILDINGS TEST PIT LOGS.GPJ TESTPIT.GDT 9/10/25

SUMMARY OF PHYSICAL PROPERTIES TEST RESULTS

PROJECT: NAVAJO GALLUP WATER SUPPLY PROJECT **FEATURE: TURNOUT BUILDINGS**

IDENTIFICATION			PARTICLE SIZE FRACTIONS IN PERCENT						CONSISTENCY LIMITS			IN-PLACE DENSITY				COMPACTION TESTS				
TEST PIT NUMBER	DEPTH – feet	CLASSIFICATION SYMBOL	FINES		SAND #200 (.074mm) to #4 (4.76mm)	GRAVEL #4 (4.76mm) to 3" (76.2mm)	COBBLES 3" (76.2mm) to 5" (127mm)	OVERSIZE Larger than 5" (127mm)	LIQUID LIMIT - %	PLASTICITY INDEX - %	SHRINKAGE LIMIT - %	DRY DENSITY - pcf	FILL WATER CONTENT - %	SPECIFIC GRAVITY PLUS No. 4	SPECIFIC GRAVITY MINUS No. 4	MAXIMUM DRY DENSITY - pcf	OPTIMUM WATER CONTENT - %	PENETRATION RESISTANCE - psi	D-VALUE - %	RELATIVE COMPACTION - %
			SMALLER THAN 0.005 mm	0.005 to0.074 mm																
*TPTB-25-1	4.5	SM	11.4	23.3	65.3	0	-	-	NP	NP	-	91.6	2.6	-	2.62	109.9	13.6	800	83.3	-
*TPTB-25-2	4.0	SM	19.1	20.9	60.0	0	-	-	NP	NP	-	86.3	3.7	-	2.63	116.9	13.3	740	73.8	-
*TPTB-25-3	5.0	SC	26.8	18.6	54.6	0	-	-	26.6	11.5	-	90.4	6.3	-	2.60	114.4	15.0	350	79.0	-
*TPTB-25-4	4.0	SM	23.8	18.5	57.7	0	-	-	NP	NP	-	91.0	6.9	-	2.63	115.1	14.1	700	79.1	-
*TPTB-25-5	5.0	SM	14.2	11.5	74.3	0	-	-	NP	NP	-	93.8	6.4	-	2.61	107.6	13.5	1050	87.2	-
*TPTB-25-6	4.5	SM	9.9	7.4	82.7	0	-	-	NP	NP	-	99.4	6.0	-	2.62	112.2	15.5	680	88.6	-
*TPTB-25-7	4.5	s(CL)	39.1	17.9	43.0	0	-	-	32.1	15.9	-	84.1	14.8	-	2.67	110.8	16.0	550	75.9	-
*TPTB-25-8	4.5	s(CL)	24.8	39.9	35.3	0	-	-	27.9	10.6	-	92.8	6.3	-	2.66	112.5	14.4	1050	82.5	-
*TPTB-25-9	4.5	SC-SM	13.5	21.6	54.5	10.4	-	-	21.6	4.8	-	94.3	6.3	2.29	2.55	117.0	11.0	1750	80.6	-
*TPTB-25-10	5.5	s(CL)	30.3	32.8	36.9	0	-	-	34.4	16.6	-	92.6	12.6	-	2.65	109.8	17.2	355	84.3	-
*TPTB-25-12	7.0	SM	9.8	10.3	79.9	0	-	-	NP	NP	-	92.3	2.4	-	2.65	114.3	10.8	900	80.7	-

** Denotes In-place density and 5-point curve*